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Thermodynamic assessment of the La–Zn system

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ABSTRACT

A thermodynamic optimization of the La–Zn system is carried out using the CALPHAD (CALculation of the PHAse Diagram) method. This new assessment takes into account the latest experimental values of enthalpies of formation reported in literature. The solution phases (liquid, bcc-La, fcc-La, dhcp-La and hcp-Zn) are modelled as substitution solutions using the Redlich–Kister equation. Nine intermediate phases are considered in this assessment, namely LaZn, $LaZn_2$, $LaZn_4$, $LaZn_5$, $LaZn_7$, La_3Zn_{22}, La_2Zn_{17}, $LaZn_{11}$ and $LaZn_{13}$ and are treated as stoichiometric compounds.

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0. Introduction

Due to their low density, magnesium–zinc alloys are very attractive structural materials for applications in the transport industry. However, their mechanical performances are weak and alloying elements are required to improve their properties. Among these elements some Rare Earth (RE) elements such as lanthanum, cerium or neodymium are very efficient [1–3]. The study of the RE-Mg–Zn systems is therefore of great interest. The optimization of specific properties requires a good knowledge of the thermochemical equilibrium and hence a good knowledge of the phase diagram.

The present work concerns the computational study of the La–Zn system. It is a part of a larger project devoted to the La–Mg–Zn system.

The La–Zn system was experimentally investigated by several authors and has already been assessed by Qi [4]. However, large discrepancies remain between the calculated and the experimental phase diagrams. In addition, enthalpies of formation recently measured by Berche [5] were not taken into account in [4]. Thus, new assessment is necessary to give a consistent set of thermodynamic data.

1. Bibliographic description of the system

1.1. The phase diagram

The phase diagram data reported in the literature [6–9] were critically reviewed in 2009 by Berche [10] showing some discrepancies between the data. New experimental investigations were performed later on by Berche [11] to solve these discrepancies and the suggested phase diagram is presented in Fig. 1. The corresponding invariant reactions are gathered in Table 1.

The invariant reactions calculated in this work are also reported in the same table and are commented in Section 3.

1.2. Thermodynamic data

Electromotive force (e.m.f.) measurements were performed in the temperature range 700–1020 K by Lesourd [12], Mullayanov [13] and Johnson [14]. The studied alloys were constituted of liquid zinc saturated with lanthanum and the measurements were referred to pure lanthanum. The composition of the alloys is only reported in [13] as varying from $x_{Zn} = 0.9963$ to $x_{Zn} = 0.9999$. These sets of data are in good agreement with each other.

Kovalevskii [15] performed e.m.f. measurements at 1000 K for 72 La–Zn alloys with compositions ranging from pure Zn to pure La. The step changes (Fig. 2, dash line) of the emf versus composition curve [15] suggests the following intermediate compounds : LaZn, $LaZn_2$, $LaZn_4$, $LaZn_6$, $LaZn_7$, $LaZn_8$, $LaZn_9$ and $LaZn_{11}$. The existence of LaZn, $LaZn_2$, $LaZn_4$, La_3Zn_{22} ($\approx LaZn_8$), La_2Zn_{17} ($\approx LaZn_9$) and $LaZn_{11}$ is supported by XRD studies as shown in the review [11]. In the interval $0.80 < x_{Zn} < 0.88$, the existence of $LaZn_6$ and $LaZn_7$ is not confirmed by other literature investigations. In this range, X-ray investigations have only shown the existence of $LaZn_5$.

In the composition range $x_{Zn} = 0.0$ to $x_{Zn} = 0.5$, at 1000 K, step changes in e.m.f. measurements must occur between the four domains: [LaZn + liquid], [liquid], [liquid + fcc-La] and [fcc-La]. However, the changes are not apparent and probably too small to be detected.

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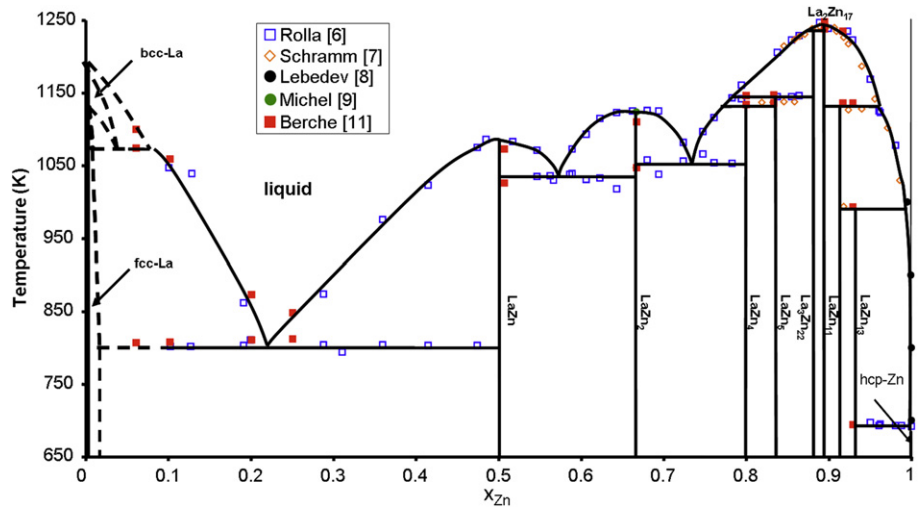

Fig. 1. The La–Zn phase diagram proposed by Berche [11].

Table 1
Invariant reactions in the La–Zn system ; n/d means no data.

Reaction (on cooling)	x_{Zn} of the phases involved			Temperature (K)	Reference
fcc-La = LaZn + dhcp-La	n/d	0.500	n/d	–	[11]
	0.004	0.500	0.000	519	This work
Liquid = fcc-La + LaZn	0.220	n/d	0.500	810 ± 2	[11]
	0.231	0.027	0.500	811	This work
bcc-La = fcc-La + liquid	n/d	n/d	n/d	1074 ± 3	[11]
	0.021	0.006	0.074	1074	This work
Liquid = LaZn	0.500	0.500	–	1085	[11]
	0.500	0.500	–	1085	This work
Liquid = LaZn + $LaZn_2$	0.565	0.500	0.667	1026 ± 2	[11]
	0.565	0.500	0.667	1031	This work
Liquid = $LaZn_2$	0.667	0.667	–	1125	[11]
	0.667	0.667	–	1123	This work
Liquid = $LaZn_2$ + $LaZn_4$	0.734	0.667	0.800	1047 ± 2	[11]
	0.731	0.667	0.800	1055	This work
Liquid + $LaZn_5$ = $LaZn_4$	n/d	0.833	0.800	1135 ± 3	[11]
	0.793	0.833	0.800	1128	This work
Liquid + La_3Zn_{22} = $LaZn_5$	n/d	0.880	0.833	1146 ± 3	[11]
Liquid + $LaZn_7$ = $LaZn_5$	0.807	0.875	0.833	1144	This work
Liquid + La_3Zn_{22} = $LaZn_7$	0.853	0.880	0.875	1225	This work
Liquid + La_2Zn_{17} = La_3Zn_{22}	n/d	0.895	0.880	1238 ± 5	[11]
	0.877	0.895	0.880	1241	This work
Liquid = La_2Zn_{17}	0.895	0.895	–	1248 ± 3	[11]
	0.895	0.895	–	1248	This work
Liquid + La_2Zn_{17} = $LaZn_{11}$	n/d	0.895	0.917	1136 ± 2	[11]
	0.956	0.895	0.917	1134	This work
Liquid + $LaZn_{11}$ = $LaZn_{13}$	n/a	0.917	0.929	993 ± 2	[11]
	0.986	0.917	0.929	991	This work
Liquid = hcp-Zn + $LaZn_{13}$	≅ 1.000	1.000	0.929	694 ± 1	[11]
	≅ 1.000	1.000	0.929	692	This work

For each two-phase regions Kovalevskii [15] measured the evolution of the e.m.f. versus temperature between 700 and 1120 K. These values are referred to saturated liquid zinc i.e. [liquid + $LaZn_{13}$] for temperatures lower than 993 K, and [liquid + $LaZn_{11}$] for temperatures higher than 993 K.

In order to compare all the e.m.f. results, a common reference state, pure lanthanum has been chosen. In this way, Kovalevskii values are corrected by using results from [12–14].

The e.m.f. data are plotted in Fig. 3.

Morishita determined the enthalpy of formation of $LaZn_{13}$ [16] and $LaZn_8$ [17] at 298 K by solution calorimetry in 5N hydrochloric acid bath. The entropy of formation of the phases $LaZn_{13}$ (-1.798 ± 0.4 J K^{-1} mol $atom^{-1}$) and $LaZn_8$ (-2.78 ± 0.4 J K^{-1} mol $atom^{-1}$) are also measured from thermal capacity measurements using the relaxation method in the temperature range 2–300 K. However, $LaZn_8$ phase does not exist, such a stoichiometry corresponds to $x_{Zn} = 0.889$ which is a two-phases alloy constitute of 41% of La_3Zn_{22} and 59% of La_2Zn_{17}.

The enthalpy of formation at 298 K was measured for LaZn, $LaZn_2$, $LaZn_4$, $LaZn_5$, La_3Zn_{22}, La_2Zn_{17}, $LaZn_{11}$ and $LaZn_{13}$ by Berche [5] using solution calorimetry in liquid tin. The enthalpy of formation at 0 K for LaZn, $LaZn_2$, $LaZn_5$ and La_2Zn_{17} was also calculated using the density functional theory (DFT). Experimental and DFT results are in agreement (Fig. 4) as discussed in [5].

1.3. Previous assessment

The La–Zn system has already been assessed by Qi [4]. The calculated phase diagram is presented in Fig. 5. Some difficulties are noticed :

– LaZn and bcc-La are described as a single phase with an order-disorder transition. As explained in [18] such a

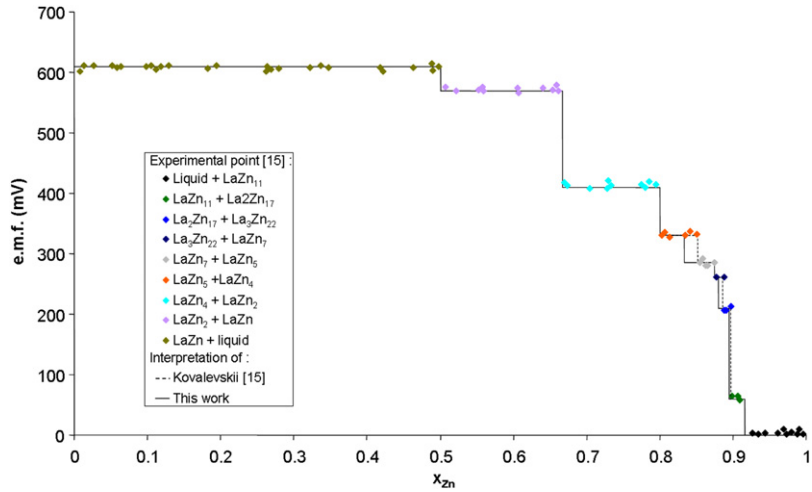

Fig. 2. E.m.f. measurements of Kovalevskii at 1000 K [15] interpreted according to Kovalevskii and to this work.

Table 2
Phase models and assessed parameters of the La–Zn system; "Va" designates vacancies.

Phase model	Thermodynamic parameters
Liquid(La, Zn)	$G(\text{liquid, La}) - {}^{\circ}H^{\text{SER}}_{\text{La}} = G^{\text{liq}}_{\text{La}}$
	$G(\text{liquid, Zn}) - {}^{\circ}H^{\text{SER}}_{\text{Zn}} = G^{\text{liq}}_{\text{Zn}}$
	$L^0_{(\text{La,Zn})} = -107900 + 9.1 * T$
	$L^1_{(\text{La,Zn})} = 78100 - 25 * T$
	$L^2_{(\text{La,Zn})} = -37100 + 17 * T$
Dhcp-La(La, Zn)	$G(\text{dhcp, La}) - {}^{\circ}H^{\text{SER}}_{\text{La}} = G^{\text{dhcp}}_{\text{La}}$
	$G(\text{dhcp, Zn}) - {}^{\circ}H^{\text{SER}}_{\text{Zn}} = G^{\text{hcp}}_{\text{Zn}} + 500$
Fcc-La(La, Zn)(Va)	$G(\text{fcc, La : Va}) - {}^{\circ}H^{\text{SER}}_{\text{La}} = G^{\text{fcc}}_{\text{La}}$
	$G(\text{fcc, Zn : Va}) - {}^{\circ}H^{\text{SER}}_{\text{Zn}} = G^{\text{fcc}}_{\text{Zn}}$
	$L^0_{(\text{La,Zn:Va})} = -44900$
Bcc-La(La, Zn)(Va)	$G(\text{bcc, La : Va}) - {}^{\circ}H^{\text{SER}}_{\text{La}} = G^{\text{bcc}}_{\text{La}}$
	$G(\text{bcc, Zn : Va}) - {}^{\circ}H^{\text{SER}}_{\text{Zn}} = G^{\text{bcc}}_{\text{Zn}}$
	$L^0_{(\text{La,Zn:Va})} = -56300$
$LaZn(La)_{0.5}(Zn)_{0.5}$	$G(\text{LaZn}) - 0.5{}^{\circ}H^{\text{SER}}_{\text{La}} - 0.5\,{}^{\circ}H^{\text{SER}}_{\text{Zn}} = -32800 + 0.06 * T + 0.5G^{\text{dhcp}}_{\text{La}} + 0.5\,G^{\text{hcp}}_{\text{Zn}}$
$LaZn_2(La)_{0.33}(Zn)_{0.67}$	$G(\text{LaZn}_2) - 0.33{}^{\circ}H^{\text{SER}}_{\text{La}} - 0.67\,{}^{\circ}H^{\text{SER}}_{\text{Zn}} = -34900 + 0.33G^{\text{dhcp}}_{\text{La}} + 0.67\,G^{\text{hcp}}_{\text{Zn}}$
$LaZn_4(La)_{0.20}(Zn)_{0.80}$	$G(\text{LaZn}_4\text{, La : Zn}) - 0.20{}^{\circ}H^{\text{SER}}_{\text{La}} - 0.80\,{}^{\circ}H^{\text{SER}}_{\text{Zn}} = -34000 + 3.63 * T + 0.20G^{\text{dhcp}}_{\text{La}} + 0.80\,G^{\text{hcp}}_{\text{Zn}}$
$LaZn_5(La)_{0.17}(Zn)_{0.83}$	$G(\text{LaZn}_5) - 0.17{}^{\circ}H^{\text{SER}}_{\text{La}} - 0.83\,{}^{\circ}H^{\text{SER}}_{\text{Zn}} = -32300 + 3.98 * T + 0.17G^{\text{dhcp}}_{\text{La}} + 0.83G^{\text{hcp}}_{\text{Zn}}$
$LaZn_7(La)_{0.125}(Zn)_{0.875}$	$G(\text{LaZn}_7) - 0.125{}^{\circ}H^{\text{SER}}_{\text{La}} - 0.875\,{}^{\circ}H^{\text{SER}}_{\text{Zn}} = -28600 + 3.3 * T + 0.125G^{\text{dhcp}}_{\text{La}} + 0.875G^{\text{hcp}}_{\text{Zn}}$
$La_3Zn_{22}(La)_{0.12}(Zn)_{0.88}$	$G(\text{La}_3\text{Zn}_{22}) - 0.12{}^{\circ}H^{\text{SER}}_{\text{La}} - 0.88\,{}^{\circ}H^{\text{SER}}_{\text{Zn}} = -28000 + 3.17 * T + 0.12G^{\text{dhcp}}_{\text{La}} + 0.88G^{\text{hcp}}_{\text{Zn}}$
$La_2Zn_{17}(La)_{0.105}(Zn)_{0.895}$	$G(\text{La}_2\text{Zn}_{17}) - 0.105{}^{\circ}H^{\text{SER}}_{\text{La}} - 0.895\,{}^{\circ}H^{\text{SER}}_{\text{Zn}} = -25700 + 2.61 * T + 0.105G^{\text{dhcp}}_{\text{La}} + 0.895\,G^{\text{hcp}}_{\text{Zn}}$
$LaZn_{11}(La)_{0.083}(Zn)_{0.917}$	$G(\text{LaZn}_{11}) - 0.083{}^{\circ}H^{\text{SER}}_{\text{La}} - 0.917\,{}^{\circ}H^{\text{SER}}_{\text{Zn}} = -21210 + 1.79 * T + 0.083G^{\text{dhcp}}_{\text{La}} + 0.917\,G^{\text{hcp}}_{\text{Zn}}$
$LaZn_{13}(La)_{0.071}(Zn)_{0.929}$	$G(\text{LaZn}_{13}) - 0.071{}^{\circ}H^{\text{SER}}_{\text{La}} - 0.929\,{}^{\circ}H^{\text{SER}}_{\text{Zn}} = -18760 + 1.66 * T + 0.071G^{\text{dhcp}}_{\text{La}} + 0.929\,G^{\text{hcp}}_{\text{Zn}}$
Hcp-Zn(Zn : Va)	$G(\text{hcp, Zn : Va}) - {}^{\circ}H^{\text{SER}}_{\text{Zn}} = G^{\text{hcp}}_{\text{Zn}}$

description is not suitable if one wants to use the calculated parameters for subsequent assessment of higher order systems.

- The zinc solubility in the various allotropic forms of lanthanum (bcc, fcc and dhcp) is assumed to be negligible. Some calculated invariant temperatures differ from the experimental ones. This difference reaches 20 K and 33 K for the [liquid = $LaZn_2$ + $LaZn_4$] and [La_3Zn_{22} + liquid = $LaZn_5$], respectively.
- Qi used the enthalpy of formation of $LaZn_{13}$ determined by Morishita [16] but the enthalpies of formation measured by Berche [5] are not taken into account.
- The electromotive forces calculated from the assessed parameters of Qi are presented in Fig. 3(a). Except for [$LaZn_2$ + LaZn], large discrepancies are observed between experimental and calculated values.

2. Thermodynamic modelling

The analytical expressions of the Gibbs energy and the parameters used in this work are gathered in Table 2.

The Gibbs energy function (G_i^{ϕ}) of pure elements (i = La or Mg) in different phase structures (ϕ) are taken from the SGTE database [19]. In this formulation, ${}^{\circ}H_i^{\text{SER}}$ corresponds to the molar enthalpy of the element i at 298.15 K in its stable element reference (SER): dhcp for lanthanum and hcp for zinc.

2.1. Solutions

In this work, liquid, bcc-La and fcc-La are modelled as substitution solutions (La, Mg). The Gibbs free energy (${}^0G_i^{\phi}$) of the phase ϕ is given by the following equation :

$$ {}^0G_i^{\phi} - \sum_{i=\text{La,Zn}} x_i\,{}^0H_i^{\text{SER}}(298.15\text{ K}) = {}^{ref}G + {}^{id}G + {}^{xs}G \quad (1) $$

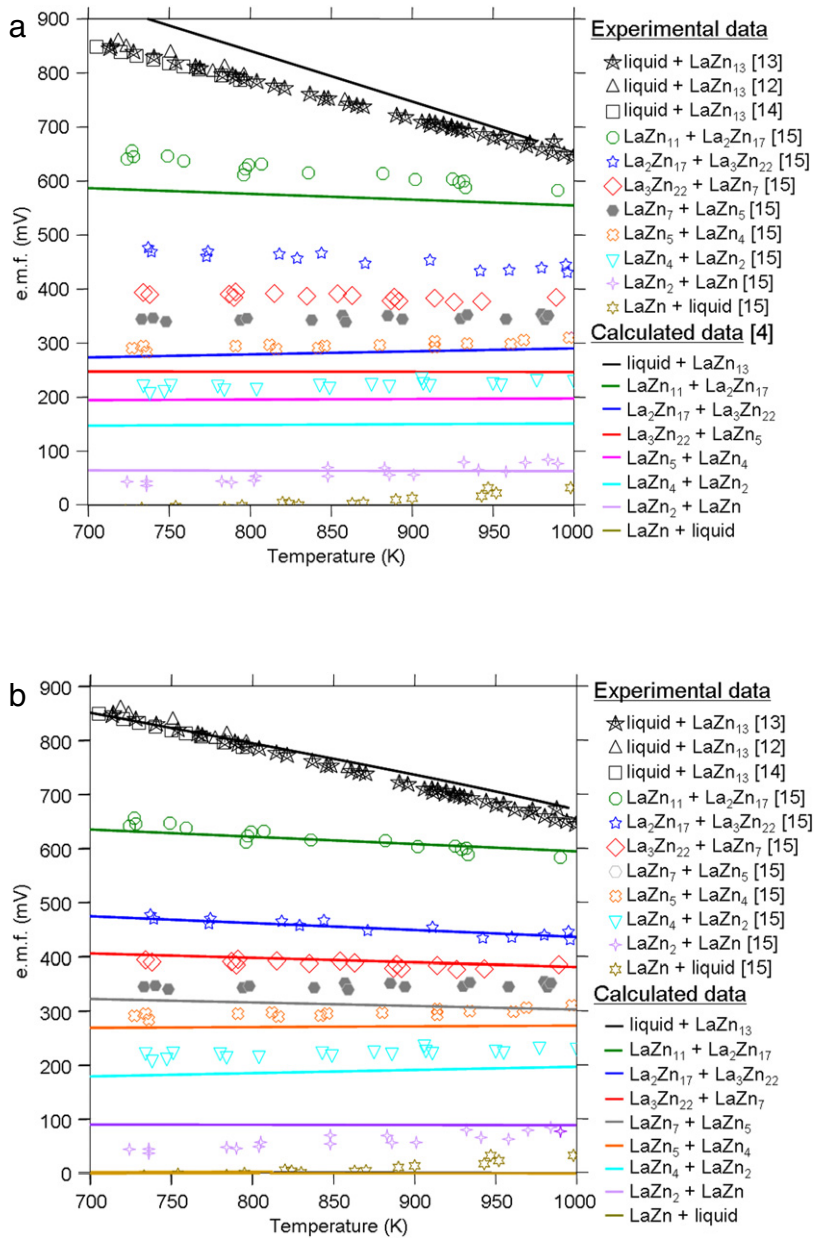

Fig. 3. E.m.f. values referred to fcc-La versus temperature between 700 and 1000 K. The experimental values are from [12–14] for liquid zinc saturated in lanthanum and from [15] for the other compositions. These experimental values are compared with the calculated ones from the assessments (a) Qi [4], (b) this work.

where the reference and the ideal terms are written as :

$$^{ref}G = \sum_{i=\mathrm{La,Zn}} x_i({}^0G_i(T) - {}^0H_i^{\mathrm{SER}}(298.15\ \mathrm{K})) \quad (2)$$

$$^{id}G = RT \sum_{i=\mathrm{La,Zn}} x_i \ln(x_i) \quad (3)$$

and the excess term is described according to the Redlich–Kister polynomial expression :

$$^{xs}G = x_{\mathrm{La}}x_{\mathrm{Zn}} \sum_n x_i L^n_{\mathrm{La,Zn}}(x_{\mathrm{La}} - x_{\mathrm{Zn}})^n \quad (4)$$

where

$$L^n_{\mathrm{La,Zn}} = a + bT. \quad (5)$$

In order to describe the solubility of zinc in dhcp-La, a positive term of 500 J is introduced in the Gibbs free energy of the metastable dhcp-Zn phase.

2.2. Intermetallic phases

All the intermediate phases are treated as line compound.

The corresponding Gibbs energy is given by the Eq. (5). Due to a lack of experimental data, the excess specific heat is supposed to be null in this description.

$$G_m^{\mathrm{La_xZn_y}} = xGHSERLA + yGHSERZN + a + bT. \quad (6)$$

3. Results and discussion

The a and b parameters in Eqs. (5) and (6) are assessed from the experimental data presented in Section 1 by using the PARROT modulus [20] of the ThermoCalc software [21].

Initially, $LaZn_7$ was not taken into account in the assessment. However it was not possible to fit the e.m.f data of Kovalevskii in the vicinity of $x_{Zn} = 0.85$.

Subsequently $LaZn_7$ was added and its decomposition was assumed to be peritectic as it is observed for La_3Zn_{22}, $LaZn_5$

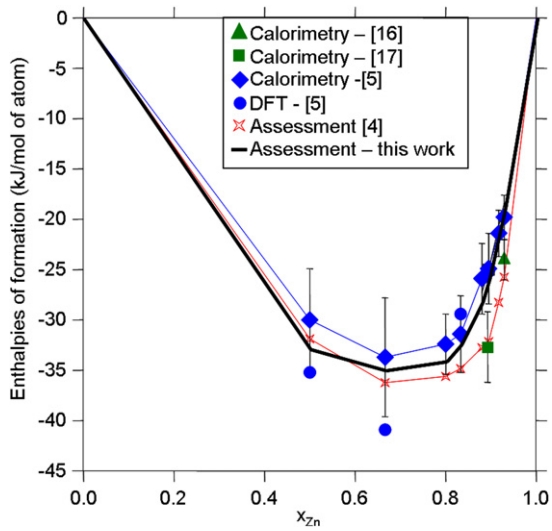

Fig. 4. Enthalpies of formation of the intermetallic compounds. Comparison between experimental [5,16,17] and calculated results.

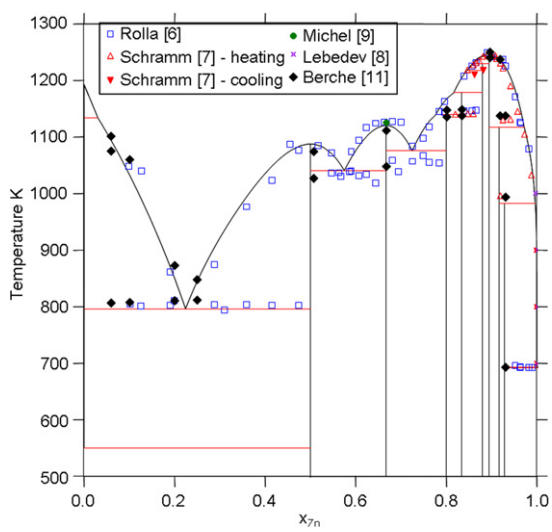

Fig. 5. Phase diagram calculated from the assessed parameters of Qi [4]. The experimental data selected by Berche [11] are plotted for comparison.

and $LaZn_4$. From this perspective, thermal arrests quoted by Schramm [7] on cooling for $x_{Zn} = 0.864$ at 1207 K and for $x_{Zn} = 0.88$ at 1212 K, and initially interpreted as the peritectic decomposition of La_3Zn_{22} with a high degree of undercooling [11], correspond now to the peritectic temperature of $LaZn_7$.

A satisfactory agreement between calculated e.m.f. and the Kovalevskii experimental results is obtained (Fig. 3(b)).

As shown in Fig. 4, the enthalpies of formation of the intermetallics phases also agree well with the experimental values of Berche [5].

The entropy of formation of $LaZn_{13}$ is calculated at -1.66 J K^{-1} mol $atom^{-1}$ at 298 K in agreement with the value of Morishita [16] depending to experimental uncertainties. The entropy of formation of a two-phases alloy for $x_{Zn} = 0.889$ is calculated at -2.834 J K^{-1} mol $atom^{-1}$ at 298 K which is also very similar to the -2.78 J K^{-1} mol $atom^{-1}$ measured by Morishita [17].

The calculated phase diagram is presented in Fig. 6.

In order to fit the temperature and the composition of the phases involved in the eutectic reactions [liquid = fcc − La + LaZn] and [liquid = $LaZn_2$ + $LaZn_4$], the L^0, L^1 and L^2 parameters of the liquid phase were made temperature dependent.

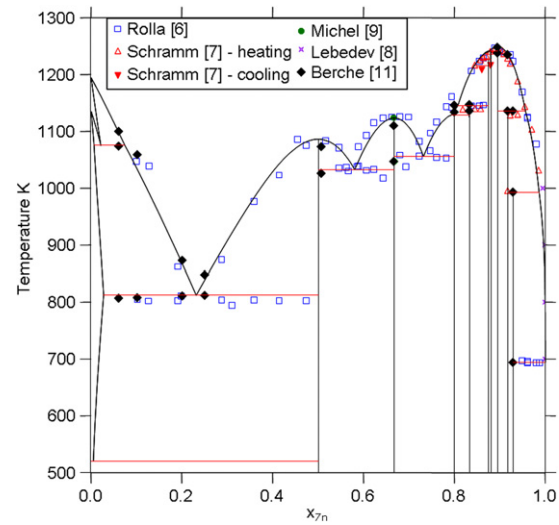

Fig. 6. The La–Zn phase diagram calculated in the present work, selected experimental data are superimposed.

The liquidus experimental points around $LaZn_2$ are not well fitted. Several hypothesis could explain this discrepancy. The fact that e.m.f. and enthalpy of formation are well reproduced in this region casts doubt on the accuracy of the experimental liquidus temperatures. These measurements could be checked by new Differential Thermal Analysis investigations. Another possibility is a strong thermal dependence of the thermodynamic properties of $LaZn_2$ which could be detected by thermal capacity measurements versus temperature for $LaZn_2$.

4. Conclusion

The La–Zn system is assessed by using the CALPHAD method. An overall good agreement is obtained between literature and calculated data.

This satisfactory description is achieved by including the $LaZn_7$ compound. Existence of this phase has still to be proven and its peritectic temperature has to be confirmed.

Experimental thermodynamic data of the liquid phase are still missing. Measurements of the enthalpy of mixing by solution calorimetry or activity by e.m.f. or vapour pressure method would be valuable. Moreover, heat capacity between 2 and 298 K could be measured systematically for each intermediate phases.

The temperatures of transition for alloys with compositions ranging from $x_{Zn} = 0.6$ to $x_{Zn} = 0.7$ need to be combined.

This work combined with the thermodynamic descriptions of La–Mg [18,22,23] and Mg–Zn [24,25] will soon be used to obtain a description of the ternary system La–Mg–Zn.

References

[1] L.Y. Wei, G.L. Dunlop, H. Westengen, The intergranular microstructure of cast Mg–Zn and Mg–Zn-rare earth alloys, Metall. Mater. Trans. A 26 (8) (1995) 1947–1955.
[2] L.Y. Wei, G.L. Dunlop, H. Westengen, Precipitation hardening of Mg–Zn and Mg–Zn–RE alloys, Metall. Mater. Trans. A 26 (7) (1995) 1705–1716.
[3] W. Wu, Y. Wang, X. Zheng, L.J. Chen, Z. Liu, Effect of neodynium on mechanical behavior of Mg–Zn–Zr magnesium alloy, J. Matter. Sci. Lett. 22 (6) (2003) 445–447.
[4] H.Y. Qi, G.X. Huang, R.D. Liu, K. Zhang, L.B. Liu, Z.P. Jin, Thermodynamic optimization of La–Zn and La–Mg–Zn systems, J. Alloys Compd. 497 (2010) 336–343.
[5] A. Berche, F. Marinelli, G. Mikaelian, J. Rogez, M.C. Record, Enthalpies of formation of the La–Zn compounds between 298 and 910 K: experimental and theoritical investigations, J. Alloy Compd. 475 (2009) 79–85.
[6] L. Rolla, A. Iandelli, Contributi alla conoscenza dei metalli e delle leghe delle terre rare, Ric. Sci. 12 (1941) 1216–1226.

[7] J. Schramm, Die systeme zink–cer und zink–lanthan, Z. Mettalkd. 33 (10) (1941) 358–360.
[8] V.A. Lebedev, I.F. Nichkov, S.P. Raspopin, R.K. Mullayanov, B.G. Semenov, Thermodynamics of the reaction of zinc with lanthanum and cerium, Russ. J. Phys. Chem. 45 (8) (1971) 1126–1127.
[9] D.J. Michel, E. Ryba, Melting points of $ReZn_2$ intermetallic compounds, J. Less Common Met. 11 (1) (1966) 67–69.
[10] A. Berche, M.C. Record, J. Rogez, Critical Review of the La–Zn system, The Open Thermodynamic Journal 3 (2009) 7–16.
[11] A. Berche, J. Rogez, M.C. Record, New experimental investigation of the Lanthanum Zinc phase diagram, Thermochim. Acta (2011), doi:10.1016/j.tca.2011.05.001.
[12] J.P.P.F Lesourd, J.A. Plambeck, Electrochemical studies of Lanthanum and Lanthanum–Zinc alloys in fused LiCl–KCl eutectic, Can. J. Chem. 47 (18) (1969) 3387–3391.
[13] R.K. Mullayanov, V.A. Lebedev, Y.P. Kanashin, I.F. Nichkov, S.P. Raspopin, Thermodynamics of the Lantahnum–Zinc system, Russ. J. Phys. Chem. 43 (11) (1969) 1559–1561.
[14] I. Johnson, R.M. Yonco, Thermodynamics of cadmium- and zinc-rich alloys in the Cd–La, Cd–Ce, Cd–Pr, Zn–La, Zn–Ce and Zn–Pr systems, Metall. Trans. 1 (4) (1970) 905–910.
[15] A.V. Kovalevskii, V.A. Lebedev, I.F. Nichkov, S.P. Raspopin, Thermodynamic properties and phase composition of La–Zn alloys, Izv. Akad. Nauk SSR Met. 1 (1972) 183–187.
[16] M. Morishita, K. Koyama, K. Tsuboki, Calorimetric study of $Zn_{13}La$, Z. Metallkd. 95 (8) (2004) 708–712.
[17] M. Morishita, H. Yamamoto, K. Tsuboki, Y. Matsumoto, Standard Gibbs energy of formation of Zn_8La determined by solution calorimetry and Measurement of heat capacity from near absolute zero Kelvin, Mater. Trans. 47 (6) (2006) 1555–1559.
[18] A. Berche, P. Benigni, J. Rogez, M.-C. Record, Thermodynamic assessment of the La–Mg system, CALPHAD: Computer Coupling of Phase Diagrams and Thermochemistry 35 (2011) 580–587.
[19] SGTE Data for Pure Elements, NPL Report DMA(A) 195, 1989.
[20] B. Jannson, Report TRITA-MAC-0234, Royal Institute Technology S10044, Stockholm, Sweden, 1984.
[21] B. Sundman, B. Jannson, J.O. Andersson, The thermo-calc databank, Calphad 9 (1985) 153–190.
[22] A. Berche, F. Marinelli, F. Rogez, M.C. Record, Enthalpy of formation of the La–Mg intermediate phases, Thermochim Acta 499 (2010) 65–70.
[23] A. Berche, P. Benigni, J. Rogez, M.C. Record, Re-investigation of the La–Mg phase diagram, J. Therm. Anal. Calo. (2012) doi:10.1007/s10973-011-1441-9.
[24] A. Berche, C. Drescher, J. Rogez, M.C. Record, S. Brühne, W. Assmus, Thermodynamic measurements in the Mg–Zn system, J. Alloy Compd. 503 (2010) 44–49.
[25] P. Liang, T. Tarfa, J.A. Robinson, S. Wagner, P. Ochin, M.G. Harmelin, H.J. Seifert, H.L. Lukas, F. Aldinger, Experimental investigation and thermodynamic calculation of the Al–Mg–Zn system, Termochim. Acta 314 (1998) 87–110.